

Table 1: Complete performance comparison at 100k steps vs SafePO at 5M steps.

Algorithm	Scenario	Score $\uparrow$	Cost $\downarrow$	Winrate $\uparrow$	Steps	Cost Limit
MACPO	1c3s5z	8.72 ± 0.68	9.00 ± 0.00	0.00 ± 0.00	5M	0
MACPO	1c3s5z	7.64 ± 0.17	9.00 ± 0.00	0.00 ± 0.00	100k	0
Safe Dreamers	1c3s5z	19.48 ± 0.35	3.40 ± 1.08	0.86 ± 0.10	100k	0
MACPO	1c3s5z	10.07 ± 1.76	9.00 ± 0.00	0.00 ± 0.00	5M	4
MACPO	1c3s5z	7.79 ± 0.05	9.00 ± 0.00	0.00 ± 0.00	100k	4
Safe Dreamers	1c3s5z	19.17 ± 0.21	4.94 ± 0.49	0.79 ± 0.03	100k	4
MACPO	2m_vs.1z	11.74 ± 8.95	1.01 ± 1.08	0.49 ± 0.54	5M	0
MACPO	2m_vs.1z	2.71 ± 0.19	2.00 ± 0.00	0.00 ± 0.00	100k	0
Safe Dreamers	2m_vs.1z	18.67 ± 0.29	0.27 ± 0.04	0.90 ± 0.02	100k	0
MACPO	2m_vs.1z	12.05 ± 8.71	1.00 ± 1.09	0.50 ± 0.55	5M	1
MACPO	2m_vs.1z	3.13 ± 0.01	2.00 ± 0.00	0.00 ± 0.00	100k	1
Safe Dreamers	2m_vs.1z	19.75 ± 0.19	0.14 ± 0.01	0.98 ± 0.02	100k	1
MACPO	2s3z	1.86 ± 1.96	5.00 ± 0.00	0.00 ± 0.00	5M	0
MACPO	2s3z	3.69 ± 0.02	5.00 ± 0.00	0.00 ± 0.00	100k	0
Safe Dreamers	2s3z	18.29 ± 0.53	2.46 ± 0.29	0.71 ± 0.09	100k	0
MACPO	2s3z	12.81 ± 7.41	3.47 ± 1.69	0.46 ± 0.51	5M	4
MACPO	2s3z	4.86 ± 0.04	5.00 ± 0.00	0.00 ± 0.00	100k	4
Safe Dreamers	2s3z	18.11 ± 0.78	2.89 ± 0.17	0.69 ± 0.11	100k	4
MACPO	2s_vs.1sc	2.51 ± 1.91	2.00 ± 0.00	0.00 ± 0.00	5M	0
MACPO	2s_vs.1sc	2.45 ± 0.12	2.00 ± 0.00	0.00 ± 0.00	100k	0
Safe Dreamers	2s_vs.1sc	19.81 ± 0.13	0.25 ± 0.09	0.95 ± 0.02	100k	0
MACPO	2s_vs.1sc	13.50 ± 7.32	1.08 ± 1.01	0.50 ± 0.54	5M	1
MACPO	2s_vs.1sc	4.21 ± 0.24	2.00 ± 0.00	0.00 ± 0.00	100k	1
Safe Dreamers	2s_vs.1sc	19.97 ± 0.12	0.32 ± 0.15	0.97 ± 0.01	100k	1
MACPO	3m	1.35 ± 1.43	3.00 ± 0.00	0.00 ± 0.00	5M	0
MACPO	3m	2.06 ± 0.12	3.00 ± 0.00	0.00 ± 0.00	100k	0
Safe Dreamers	3m	19.32 ± 0.53	0.78 ± 0.14	0.95 ± 0.04	100k	0
MACPO	3m	11.81 ± 8.96	1.64 ± 1.50	0.50 ± 0.55	5M	1
MACPO	3m	2.55 ± 0.04	3.00 ± 0.00	0.00 ± 0.00	100k	1
Safe Dreamers	3m	19.12 ± 0.42	0.81 ± 0.15	0.93 ± 0.03	100k	1
MACPO	3s5z_vs.3s6z	2.44 ± 1.10	8.00 ± 0.00	0.00 ± 0.00	5M	0
MACPO	3s5z_vs.3s6z	3.47 ± 0.03	8.00 ± 0.00	0.00 ± 0.00	100k	0
Safe Dreamers	3s5z_vs.3s6z	12.89 ± 0.67	7.81 ± 0.25	0.06 ± 0.08	100k	0
MACPO	3s5z_vs.3s6z	5.23 ± 0.27	8.00 ± 0.00	0.00 ± 0.00	5M	4
MACPO	3s5z_vs.3s6z	3.86 ± 0.04	8.00 ± 0.00	0.00 ± 0.00	100k	4

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Algorithm	Scenario	Score $\uparrow$	Cost $\downarrow$	Winrate $\uparrow$	Steps	Cost Limit
Safe Dreamers	3s5z_vs_3s6z	11.31 $\pm$ 0.16	7.99 $\pm$ 0.02	0.00 $\pm$ 0.00	100k	4
MACPO	3s_vs_3z	4.05 $\pm$ 3.23	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	0
MACPO	3s_vs_3z	6.04 $\pm$ 0.05	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
Safe Dreamers	3s_vs_3z	19.83 $\pm$ 0.29	0.89 $\pm$ 0.22	0.95 $\pm$ 0.03	100k	0
MACPO	3s_vs_3z	13.07 $\pm$ 6.56	1.91 $\pm$ 1.31	0.41 $\pm$ 0.47	5M	1
MACPO	3s_vs_3z	6.28 $\pm$ 0.11	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	1
Safe Dreamers	3s_vs_3z	19.56 $\pm$ 0.15	1.16 $\pm$ 0.32	0.92 $\pm$ 0.01	100k	1
MACPO	3s_vs_4z	3.76 $\pm$ 1.29	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	0
MACPO	3s_vs_4z	4.39 $\pm$ 0.04	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
Safe Dreamers	3s_vs_4z	20.39 $\pm$ 0.64	1.68 $\pm$ 0.39	0.83 $\pm$ 0.11	100k	0
MACPO	3s_vs_4z	6.09 $\pm$ 1.45	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	1
MACPO	3s_vs_4z	4.55 $\pm$ 0.01	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	1
Safe Dreamers	3s_vs_4z	19.68 $\pm$ 1.29	2.01 $\pm$ 0.38	0.67 $\pm$ 0.20	100k	1
MACPO	3s_vs_5z	2.89 $\pm$ 0.93	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	0
MACPO	3s_vs_5z	3.44 $\pm$ 0.03	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
Safe Dreamers	3s_vs_5z	19.81 $\pm$ 1.25	2.34 $\pm$ 0.51	0.44 $\pm$ 0.28	100k	0
MACPO	3s_vs_5z	5.21 $\pm$ 1.56	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	1
MACPO	3s_vs_5z	3.52 $\pm$ 0.02	3.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	1
Safe Dreamers	3s_vs_5z	17.42 $\pm$ 1.39	2.74 $\pm$ 0.14	0.19 $\pm$ 0.10	100k	1
MACPO	8m	0.75 $\pm$ 0.62	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	0
MACPO	8m	1.43 $\pm$ 0.01	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
Safe Dreamers	8m	19.01 $\pm$ 0.79	3.25 $\pm$ 1.14	0.88 $\pm$ 0.10	100k	0
MACPO	8m	1.37 $\pm$ 0.58	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	4
MACPO	8m	1.63 $\pm$ 0.01	8.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	4
Safe Dreamers	8m	16.75 $\pm$ 1.29	5.25 $\pm$ 0.95	0.64 $\pm$ 0.13	100k	4
MACPO	bane_vs_bane	19.72 $\pm$ 0.31	14.97 $\pm$ 7.87	0.90 $\pm$ 0.10	5M	0
MACPO	bane_vs_bane	19.26 $\pm$ 0.01	22.55 $\pm$ 0.02	0.79 $\pm$ 0.02	100k	0
Safe Dreamers	bane_vs_bane	19.13 $\pm$ 1.42	11.33 $\pm$ 9.60	0.80 $\pm$ 0.32	100k	0
MACPO	bane_vs_bane	19.64 $\pm$ 0.26	16.96 $\pm$ 6.11	0.89 $\pm$ 0.08	5M	4
MACPO	bane_vs_bane	19.37 $\pm$ 0.10	22.40 $\pm$ 0.07	0.80 $\pm$ 0.02	100k	4
Safe Dreamers	bane_vs_bane	19.75 $\pm$ 0.41	10.88 $\pm$ 6.59	0.94 $\pm$ 0.09	100k	4
MACPO	MMM	1.11 $\pm$ 0.72	10.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	0
MACPO	MMM	1.87 $\pm$ 0.01	10.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	0
Safe Dreamers	MMM	17.24 $\pm$ 0.89	7.32 $\pm$ 0.64	0.58 $\pm$ 0.06	100k	0
MACPO	MMM	1.16 $\pm$ 0.67	10.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	4
MACPO	MMM	1.89 $\pm$ 0.04	10.00 $\pm$ 0.00	0.00 $\pm$ 0.00	100k	4
Safe Dreamers	MMM	15.16 $\pm$ 1.24	7.96 $\pm$ 0.93	0.41 $\pm$ 0.13	100k	4